

# of Questions: 30

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**Question #:** 1

A 26-year-old woman comes to the physician because of a 2 week history of severe pelvic pain and urinary discomfort. Patient history shows multiple sexual partners and past treatment for gonorrhea. Pelvic examination shows pain on [palpation of the uterus. Ultrasound imaging shows uterine linings consistent with her menstrual phase. Hysterosalpingogram shows narrowing of the uterine tube. A diagnosis of salpingitis is made. Which of the following lymph nodes will most likely be enlarged as a result?

- A. Superficial and deep inguinal lymph nodes
  - B. External iliac nodes
  - C. Para-aortic nodes at the level of the renal vessels
  - D. Node of Cloquet
  - E. Internal iliac nodes accompanying the uterine artery and vein
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**Question #:** 2

A 23-year-old woman comes to the emergency department because of a 4-hour history of severe colicky pain in her right lower abdomen. She reports a last menstrual period at approximately 8-weeks earlier but denies pregnancy due to being on oral contraceptives. Physical examination shows a pale woman in clear painful distress. Palpation of the right iliac region produces sharp pain without rebound tenderness. A urine dipstick test shows presence of hCG. Transabdominal ultrasound shows an empty uterine cavity with an enlarged right uterine tube and fluid accumulation in the pelvic cavity. Culdocentesis is performed to access the nature of the fluid. Which of the following routes is most likely used to perform the culdocentesis?

- A. Perineal body into the vesicouterine space
- B. Posterior part of the vaginal fornix into the rectouterine pouch
- C. Anterior part of the vaginal fornix into the endocervical canal
- D. Vaginal orifice into the greater vestibular gland

E. Perineal membrane into the external urethral sphincter

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### Question #: 3

A 35-year-old pregnant woman comes to the obstetrician for a scheduled ultrasound exam. Physical examination shows lower than expected fundal height for her gestational time. Ultrasonography of the uterus shows fetal growth restriction, oligohydramnios and signs of abdominal distention of the fetus. Labor is induced and postpartum examination of the neonate shows no separation between the vaginal, rectal and colonic structures. Which of the following is the most likely diagnosis?

- A. Labioscrotal folds
  - B. Persistent cloaca
  - C. Urogenital folds
  - D. Genital tubercle
  - E. Urogenital membrane
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### Question #: 4

A 34-year-old woman in active labor is brought to the emergency department. Physical examination shows ruptured membranes and 4cm dilated cervix. During the delivery the fetus does not pass through the vaginal canal as expected. During inspection of the fetal path it is noted that the umbilical cord is wrapped around the fetal arm. Assisted delivery with episiotomy is done. Prior to the episiotomy being performed the physician performs a nerve block on the area. Which of the following best describes the block procedure?

- A. Injection near ischial tuberosity anesthetizes the mons pubis and anterior labia majora
  - B. Injection near ischial tuberosity anesthetizes the mons pubis and anterior labia majora
  - C. Injection near ischial spines prevents pain from the uterus, cervix and upper vagina
  - D. Injection near ischial spines anesthetizes the posterior labia majora and inferior vagina
  - E. Injection superomedial to the anterior superior iliac spine anesthetizes most perineal skin
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### Question #: 5

A 62-year-old nulliparous woman comes to the physician because of an unexpected weight loss and 2 week history of abdominal and pelvic pain. Physical examination shows tenderness on palpation of the left iliac region with a hard palpable mass. Biopsy of the mass shows dysplastic tissue consistent with epithelioid ovarian cancer. Surgical removal of the mass is planned with sentinel node removal. Which of the following lymph nodes will most likely be removed during the surgery?

- A. Deep inguinal
  - B. Superficial inguinal
  - C. Lumbar
  - D. External iliac
  - E. Internal iliac
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#### Question #: 6

A 16-year-old boy is brought to the physician by his parents because he does not appear to have started puberty. Physical examination shows a boy with less than expected musculoskeletal development. Pelvic examination shows scarce pubic hair, small penis and an empty scrotal sac. Laboratory tests show lower than expected testosterone levels. Which of the following diagnoses is most likely based on the patients condition?

- A. Congenital deficiency of 5-alpha reductase
  - B. Abdominal undescended testes resulting in cellular degradation
  - C. Absence of the receptors necessary for testosterone action
  - D. Failure of the gonads to develop
  - E. Regression of the mesonephric duct during fetal development
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#### Question #: 7

A 21-year-old woman comes to the gynecologist for her first appointment because she has recently become sexually active. Physical examination shows normal development. Pelvic examination is unremarkable. A transabdominal ultrasound is done showing bilateral ovaries but absence of the uterine tube on the left and a smaller than expected uterine cavity. Which of the following structures most likely failed to develop resulting in her condition?

- A. paramesonephric duct
  - B. mesonephric duct
  - C. mesonephros
  - D. metanephros
  - E. gonadal ridge
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### Question #: 8

A 3-year-old boy is brought to the physician by his mother because of a 1-month history of poor feeding and disinterest in playing. He is the fourth child in a family of six living in a war-torn region. Dietary history reveals that he is primarily fed a corn-based diet and often eats leftovers from meals with his older siblings. Physical examination shows an irritable patient with light-colored, friable hair, dermatitis, and mild bilateral pitting ankle edema. His weight is 70% of the expected weight for age. Which of the following is the most likely diagnosis?

- A. Cachexia
  - B. Marasmus
  - C. Kwashiorkor
  - D. Stunting
  - E. Anorexia nervosa
  - F. Rickets
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### Question #: 9

A 6-month-old infant girl is brought to the physician for a routine well-child visit. She has been exclusively breastfed since birth. Since her previous visit, she has gained 2.7 kg (6 lb). She is at the 80th percentile for weight and height. Which of the following best describes the nitrogen metabolism in this patient?

- A. She is in a state of nitrogen equilibrium
- B. Nitrogen intake is more than nitrogen output
- C. Nitrogen intake is less than nitrogen output
- D. She is in negative nitrogen balance
- E. There is excessive muscle proteolysis

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**Question #: 10**

A research group develops a formula for newborn infants whose mothers are unable to breastfeed. The formula is designed to mimic the composition of human breast milk and provide essential nutrients for optimal growth and development. Which of the following nutrients is naturally present in low concentration in human milk and should not be excessively increased in formula?

- A. Lactose
- B. Iron
- C. Docosahexaenoic acid (DHA)
- D. Leucine
- E. Human milk oligosaccharides (HMO)

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**Question #: 11**

A 4-year-old boy is brought to the physician by his parents for a routine well-child visit. His height and weight are below the average for children of his age. He performs well in school and has normal cognitive function. The parents report that he has been a “picky eater” since age 2, consistently avoiding eggs, meat, and beans. Physical examination shows normal skin and hair, no edema, and he appears active and playful. Which of the following best describes the underlying nutritional condition in this patient?

- A. Childhood obesity
- B. Stunting
- C. Cachexia
- D. Kwashiorkor
- E. Marasmus

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**Question #: 12**

A 6-month-old girl is brought to the physician for a routine well-child visit and immunizations. Her weight and length are appropriate for her age, and her developmental milestones are consistent with her age. The mother reports

exclusive breastfeeding since birth and plans to continue breastfeeding until the child is 1 year old. She is seeking advice to begin introducing semi-solid foods. Which of the following nutritional recommendations is most appropriate for this infant?

- A. Continue exclusive breastfeeding until 1 year of age
  - B. Discontinue breastfeeding and start formula feeding
  - C. Begin iron-containing complementary foods while continuing breastfeeding
  - D. Start complementary feeding with iodine supplements
  - E. Start complementary feeding with calcium supplements
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**Question #: 13**

A 30-year-old woman in the first trimester of pregnancy comes to the physician for a routine prenatal visit. Her BMI is 22 kg/m<sup>2</sup>. She has no personal or family history of major medical illnesses and takes no medications. Physical examination is unremarkable. Which of the following nutritional recommendations is most appropriate for this patient at this stage of pregnancy?

- A. An additional 2000 Cals are required from the second semester onwards
  - B. Iron supplementation prevents the occurrence of neural tube defects in the fetus
  - C. Alcohol consumption during pregnancy is associated with fetal macrosomia
  - D. Maternal iodine deficiency increases the risk of anemia in the newborn
  - E. Folic acid requirements increase during pregnancy
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**Question #: 14**

A 1-month-old infant assigned male at birth is brought to the physician for a follow-up examination. Physical examination shows ambiguous genitalia with micropenis and incomplete virilization of the external genitalia. The testes are descended bilaterally. Targeted genetic testing identifies a pathogenic mutation in the gene encoding 5-alpha reductase. Which of the following processes is most likely defective in this patient?

- A. Conversion of testosterone to dihydrotestosterone

- B. Conversion of testosterone to estrogen
  - C. Formation of T3 from T4
  - D. Formation of 1, 25 dihydroxycholecalciferol from vitamin D
  - E. Conversion of pregnenolone to dehydroepiandrosterone
  - F. Formation of cortisol
  - G. Formation of aldosterone
  - H. Formation of epinephrine from tyrosine
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**Question #: 15**

A 29-year-old woman, G1P0, comes to the physician at 11 weeks' gestation for routine prenatal testing. She has no significant medical history and takes only prenatal vitamins. A noninvasive prenatal screening test shows an increased proportion of DNA fragments derived from chromosome 21, raising concern for fetal trisomy 21. The patient is counseled about confirmatory diagnostic options. Which of the following maternal samples was used for this screening test?

- A. Amniotic fluid obtained by amniocentesis
  - B. Chorionic villi obtained by transcervical biopsy
  - C. Percutaneous umbilical cord
  - D. Paternal urine sample
  - E. Maternal blood draw
  - F. Paternal blood draw
  - G. Maternal urine sample
  - H. Cervical mucus
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**Question #: 16**

A 6-month-old infant is brought to the physician by his parents because of concerns about abnormal limb development. Physical examination shows shortened proximal limbs, an enlarged head with frontal bossing, and midface hypoplasia. Radiographs reveal abnormal organization of cartilage growth plates with disordered chondrocyte architecture. Genetic testing identifies a mutation in the FGFR3 gene. Which of the following terms best describes the underlying abnormality in this patient?

- A. Sequence
- B. Association

- C. Dysplasia
  - D. Deformation
  - E. Disruption
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**Question #: 17**

A 28-year-old woman, G2P0, goes to her physician for prenatal counseling. Her previous pregnancy resulted in an infant with severe holoprosencephaly who died shortly after birth. She denies teratogenic exposures during that pregnancy. Physical examination reveals a single maxillary central incisor. Fetal ultrasound in the current pregnancy is normal so far. Which of the following genes is most likely affected by a loss-of-function mutation in this patient?

- A. *SHH*
  - B. *FGFR3*
  - C. *CFTR*
  - D. *SHOX*
  - E. *MYC*
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**Question #: 18**

A 25-year-old woman delivers a healthy-appearing newborn boy at term without complications. On day 2 of life, routine newborn screening reveals markedly elevated blood phenylalanine levels. The infant is started on dietary therapy while confirmatory testing is arranged. The mother's pregnancy and delivery were unremarkable, and there is no family history of metabolic disease. Which of the following diagnostic tools was most likely used to detect this abnormality?

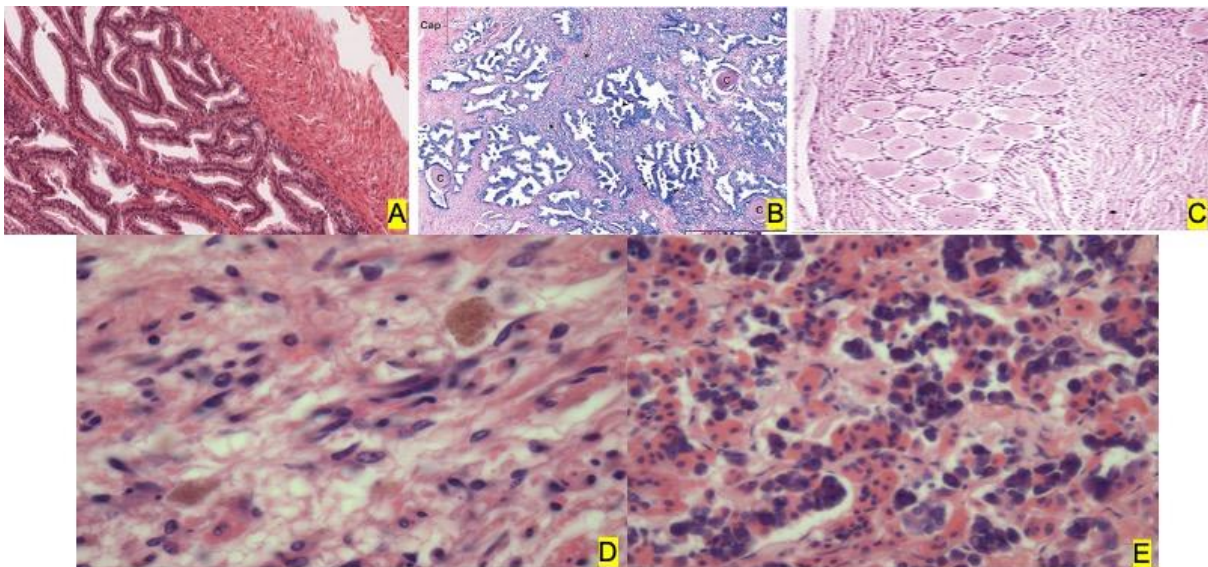
- A. Mass spectroscopy
  - B. cDNA microarray
  - C. Karyotype
  - D. Decreased AFP
  - E. Southern blot
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**Question #: 19**



A 32-year-old woman is brought to the emergency department because of a 1-hour history of vaginal bleeding. She is at 40 weeks' gestation. Physical examination shows the absence of uterine contractions. The physician suspects a deficiency of the hormone responsible for the promotion of uterine contraction during delivery. Which of the following photomicrographs best corresponds to a structure most likely affected to result in this hormonal deficiency?

- A. A
- B. B
- C. C
- D. D
- E. E




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**Question #: 20**

A 39-year-old man comes to the physician because of a 1-year history of inability to conceive. His vital signs are within normal limits. Physical examination shows no abnormalities. A biopsy of his testes show distorted morphology of a columnar shaped cell with a pale nucleus and prominent nucleolus. Which of the following is most likely a laboratory finding in this patient's condition?

- A. Low testosterone
- B. Low LH
- C. Elevated inhibin

- D. High FSH levels
  - E. Elevated ABP levels
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**Question #: 21**

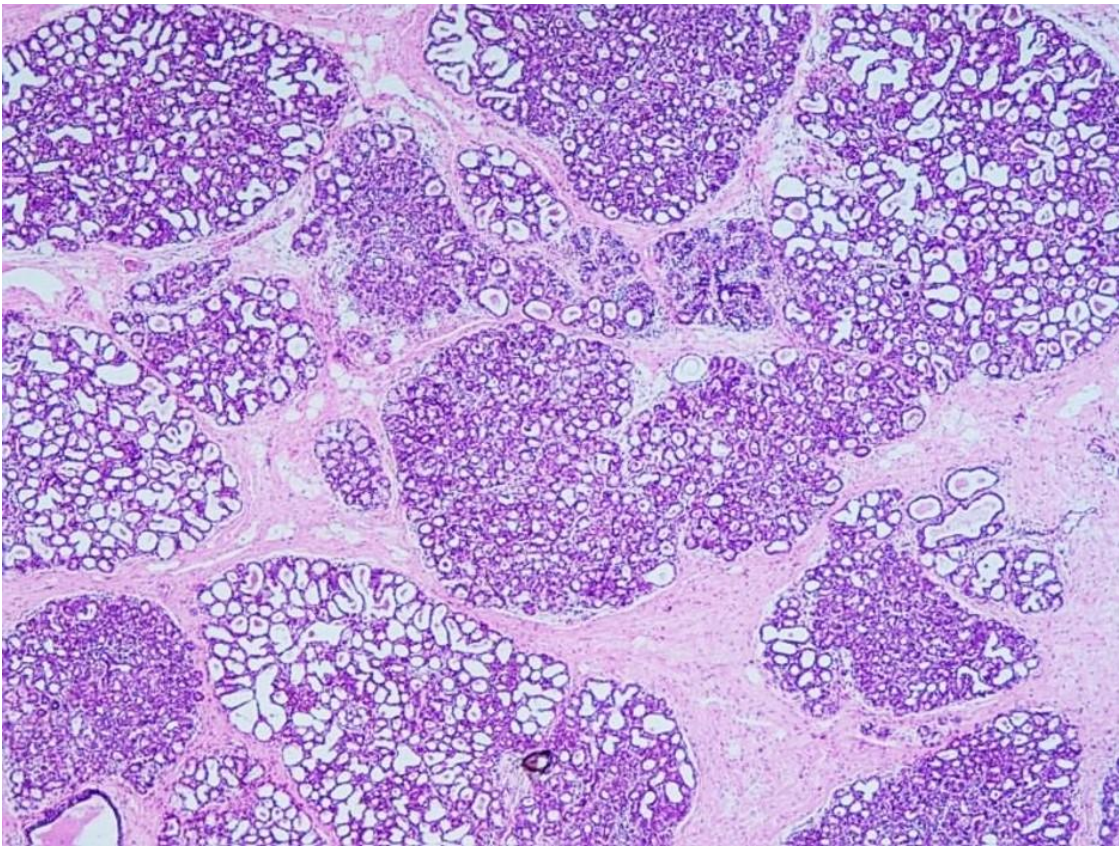
A 24-year-old woman is brought to the emergency department by her husband because of a 1-week history of intermittent vaginal bleeding. She is at 12 weeks' gestation. Physical examination shows abdominal tenderness. Ultrasound shows an intrauterine mass. Histopathological examination of a biopsy specimen shows proliferation of cytotrophoblasts and syncytiotrophoblasts deep into the endometrium. Which of the following hormones is most likely being produced by these trophoblastic cells?

- A. Prolactin
  - B. Oxytocin
  - C. hCG
  - D. Melatonin
  - E. Cortisol
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**Question #: 22**

A 32-year-old woman comes to the physician because of a 2-day history of a lump in her right breast. She is 6 months postpartum, and has stopped breast feeding 2 months prior. Physical examination shows a well-defined, hypermobile 2.5 cm x 2.5 cm mass. A biopsy specimen is shown in the attached photomicrograph. Which of the following will most likely be a normal finding on biopsy in this patient?

- A. Lots of stromal tissue
- B. Well-developed glands
- C. Underdeveloped glands
- D. Glandular atrophy
- E. Very little amount of adipocytes



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**Question #:** 23

A 28-Year-old-man comes to the physician because of a 1-year history of infertility. He was diagnosed with lymphoma and received chemotherapy 18 months ago. His vital signs are within normal limits. Physical examination shows normal male secondary sex characters, normal testes and penis. Laboratory studies show low sperm count but normal serum testosterone level. He is suspected of having Sertoli cell dysfunction. Which of the following serum hormonal alterations are most likely to be seen in FSH and LH levels respectively?

- A. Decrease, decrease
- B. Normal, normal
- C. Normal, increase
- D. Increase, normal
- E. Increase, increase



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**Question #: 24**

A 31-year-old woman at 28 weeks' gestation comes to the physician for a routine examination. This is her first antenatal visit. Her pulse is 82/min, blood pressure is 128/84 mm Hg, and respirations are 16/min. Physical examination shows normal fetal heart sounds. Laboratory studies show elevated plasma estrogen, progesterone, and prolactin levels. Which of the following is most likely decreased during this stage of gestation?

- A. Hematocrit
- B. Blood volume
- C. Fibrinogen
- D. Cardiac output
- E. Tidal volume

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**Question #: 25**

A 31-year-old woman at 24 weeks' gestation comes to the physician for a routine examination. She feels good and does not have any complaints. Her pulse is 86/min, blood pressure is 130/85 mm Hg, and respirations are 18/min. Physical examination shows normal height of the uterine fundus for the gestational age and normal fetal heart sounds. She tells the physician that she would like to know more about the placental functions. Which of the following is true regarding the function of the placenta?

- A. Allows passage of all immunoglobins
- B. It is a perfect barrier that prevents infections from transmitting to the fetus
- C. It has strong aromatase activity, which converts androgens to estrogens
- D. It is the major source of progesterone during the first six weeks of pregnancy
- E. It produces human placental lactogen (HPL), which increases maternal Insulin sensitivity

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**Question #: 26**

A 16-year-old girl is brought to the clinic by her mother for a routine examination.

Her menarche occurred at the age of 11 years. Her temperature is 37°C (98.6°F), pulse is 86/min, blood pressure is 118/ 75 mm Hg, and respirations are 16/min. Physical examination of the breast shows that the papilla and the areola project to form a secondary mound above the rest of the breast tissue. Her pubic hair resembles adult type and is distributed in the classic female triangular pattern and extends on her medial thighs. Which of the following tanner stages of puberty is this patient most likely in?

- A. Stage 5 for both breast and pubic hair development
  - B. Stage 4 for both breast and pubic hair development
  - C. Stage 4 for breast development and stage 5 for pubic hair development
  - D. Stage 4 for pubic hair development and stage 5 for breast development
  - E. Stage 3 for breast development and stage 5 for pubic hair development
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#### Question #: 27

A 47-year-old woman comes to the physician because of a 1-year history of missed periods and episodes of anxiety and irritability. Her menses previously occurred at a regular 28-day interval. She is sexually active with one male partner and uses condoms consistently. She is 163 cm (5 ft 4 in) tall and weighs 80 kg (176 lb); BMI is 30.1 kg/m<sup>2</sup> (Normal: 19-25 kg/m<sup>2</sup>). Her vital signs are within normal limits. Physical examination shows vaginal dryness and atrophy of the vaginal mucosa. Elevated serum level of which of the following hormones most likely confirms the diagnosis of menopause in this patient?

- A. Adrenal androgens
  - B. Follicle-stimulating hormone
  - C. Human chorionic gonadotropin
  - D. Adrenocorticotrophic hormone
  - E. Inhibin
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#### Question #: 28

A 39-year-old woman comes to the physician because of a 3-month history of missed periods. She is wondering if she is menopausal. She also states she has breast tenderness. She is married and has 2 children. Her last child was born at age 34. Her vital signs are within normal limits and physical examination shows no abnormalities. Her urine pregnancy test is positive and her blood tests shows high levels of prolactin, estrogen and progesterone. The action of high prolactin on milk

production in the breast tissue is most likely suppressed by which of the following hormones in this patient?

- A. Oxytocin
  - B. Estrogen and progesterone
  - C. Gonadotropin releasing hormone
  - D. Luteinizing hormone
  - E. Follicular stimulating hormone
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**Question #: 29**

A 24-year-old woman comes to the physician because of a 2-month history of missing menstrual periods. She says that she was taking birth control pills over a year ago but stopped about 6 months ago. Her vital signs are within normal limits. Physical examination shows no abnormalities. The physician suspects that she may be pregnant and requests a pregnancy test. Which of the following laboratory findings are most likely to confirm the suspected diagnosis?

- A. Low plasma progesterone and high luteinizing hormone
  - B. High prolactin, low luteinizing hormone, and low progesterone
  - C. High urinary estradiol and low progesterone
  - D. High urinary human chorionic gonadotropin and high plasma progesterone
  - E. High follicle stimulating, luteinizing hormone and human chorionic gonadotropin
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**Question #: 30**

A 25-year-old woman comes to the physician for a routine examination. She got married recently and wishes to have a child. Menarche occurred at the age of 12 years and menses occur at regular 28-day intervals. She is not on any birth control pills and is not using any other methods of contraception. She takes no medication. Physical examination shows no abnormalities. Laboratory studies show a very high serum estrogen level with low baseline serum progesterone levels. Based on these laboratory results, which of the following events is most likely to occur within the next 2 days in this woman?

- A. Cessation of menstruation

- B. Decrease in basal body temperature
- C. Ovulation
- D. Thickening of cervical mucus
- E. Regression of corpus luteum